TAWHID CHOWDHURY

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RESEARCH INTERESTS

Reinforcement learning for sequential decision-making; deep Q-networks and classical Q-learning for dynamic resource allocation; multi-agent scheduling; simulation of urban and combinatorial systems (SUMO); and applied machine learning for structured prediction from clinical and document data.

EDUCATION

BRAC University, Dhaka, Bangladesh January 2022 – March 2026
B.Sc. in Computer Science and Engineering *CGPA **3.73 /4.00***
Vice Chancellor's List (1 semester); Dean's List (6 semesters); merit-based tuition scholarship, 2022–2026.

St. Joseph Higher Secondary School, Dhaka July 2018 – December 2020
Higher Secondary Certificate, Science *GPA **5.00 / 5.00***

Dhanmondi Government Boys' High School, Dhaka January 2008 – March 2018
Secondary School Certificate, Science *GPA **5.00 / 5.00***

RESEARCH EXPERIENCE

Adaptive Urban Traffic Control via Reinforcement Learning February 2026
Undergraduate thesis · Python, SUMO, Q-learning, DQN *Thesis group leader*

Supervisors: Prof. Amitabha Chakrabarty and Prof. Md. Golam Rabiul Alam, Department of CSE, BRAC University.

- Formulated dynamic signal control as a sequential decision problem and modeled volatile urban networks in the SUMO traffic simulator.
- Designed, tuned, and compared tabular Q-learning and Deep Q-Networks against fixed-time signal heuristics, with emphasis on reward design.
- In simulated congestion scenarios, reduced average vehicle waiting time by 60–70% and increased throughput by 35–40% relative to fixed-time control.
- Led the student research team for more than a year, coordinating experiments, implementation, and supervisor milestones. **Manuscript based on this work is in preparation for peer-reviewed submission.**

SELECTED PROJECTS

Biomarker-Based Pancreatic Cancer Diagnosis | Source Code| Report January 2026
Python, scikit-learn, XGBoost, pandas *Academic Group project*

- Took a leading technical role in developing a non-invasive classification pipeline using 590 clinical and urinary biomarker samples, comparing SVM, Random Forest, and XGBoost.
- Handled missing values and class imbalance; evaluated models using precision, recall, F1, and ROC. Achieved a best accuracy of 90.68% with XGBoost.

Bangladesh NID Extractor | Source Code August 2026
FastAPI, Google Cloud Vision, OpenCV, Pillow, Docker *Independent Project*

- Built a document-understanding pipeline that extracts structured fields from Bangladesh National ID images using OCR and neural machine translation.
- Added OpenCV/Pillow quality checks and a Dockerized inference API with JSON outputs and in-memory processing (no persistent image storage).

Nirapod: All-in-One Fast Response Service | Source Code | Live May 2025
Spring Boot, PostgreSQL, JavaScript · team of four *Team lead*

- Proposed and led a modular civic-response system covering fire service, animal welfare, and city corporation workflows.
- Implemented REST APIs for complaint CRUD and filtering. Ranked among the top two projects in the course section.

Health Guardian | **Source Code** February 2024

HTML, CSS, PHP, MySQL · database systems project *Project lead*

- Designed ER/EER schemas and a symptom-to-diagnosis feature; wrote SQL for filtering, aggregation, and patient and ambulance reporting.

Embedded and graphics coursework 2023 – 2024

Arduino, Python, OpenGL

- Obstacle-avoiding line-following robot on Arduino with proximity sensing (about 10–15% fewer collisions in physical tests). 2D game rendered from OpenGL points to limit primitive overhead.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, JavaScript, Assembly, Verilog.
- **Machine Learning:** Reinforcement Learning (Q-Learning, DQN), Supervised Learning (SVM, Random Forest, XGBoost), scikit-learn.
- **Simulation & Computer Vision:** SUMO, OpenCV, Pillow, Google Cloud Vision API.
- **Development & Tools:** FastAPI, Spring Boot, REST APIs, PostgreSQL, MySQL, Docker, Git, LaTeX.
- **Hardware:** Raspberry Pi, Arduino.

HONORS AND AWARDS

- Vice Chancellor's List, BRAC University — highest academic distinction for exceptional GPA (1 semester).
- Dean's List, BRAC University — 6 semesters.
- Merit-based scholarship, BRAC University, 2022–2026.
- Physics Brawl, Open Category — ranked 104th globally; team lead, Physics Olympiad.
- Mathematics Olympiad recognition and perfect attendance, St. Joseph Higher Secondary School.

LEADERSHIP AND SERVICE

- **Organization and Management Executive, Bizbee Business Club, BRAC University (2022–2026).** Ran recruitment for the department and coordinated logistics for campus business events.
- **Joint Secretary, author, and editorial member, Josephites Math Club (2018–2020).** Managed logistics for a digital presentation competition (500+ participants) and the editorial pipeline for the annual Summation magazine.
- **Student Cabinet leader, Dhanmondi Government Boys' High School (2015–2018).** Elected to represent 800+ students. Founded the school Rubik's Cube Club and designed training sessions for 20+ members.

Internship

IT Intern October 26 - January 27

- Working as an IT intern at Sonali Bank PLC(Largest state owned commercial bank, Bangladesh)

CERTIFICATIONS

- Fully Automated MLOps;
- Developing Machine Learning Models for Production;
- MLOps Deployment and Life Cycle — DataCamp, September 2025.

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic 3 October 2026

- Overall band **8.0**. Listening 8.5; Reading 8.0; Writing 7.5; Speaking 7.5. (No band less than 7.5)
- Other languages: Bangla (native); Hindi (conversational).

REFERENCES

Dr. Amitabha Chakrabarty, Professor, Department of CSE, BRAC University (Thesis supervisor) | Email

Dr. Md. Golam Rabiul Alam, Professor, Department of CSE, BRAC University (Thesis supervisor) | Email

Dr. Swakkhar Shatabda, Professor, Department of CSE, BRAC University | Email